

# ABOLFAZL (ARASH) SODAGARTOJGI

🎓 PhD Candidate in Statistics, Rutgers University  
✉ [sodagaax@gmail.com](mailto:sodagaax@gmail.com) 📞 (540) 607-4507 📍 Highland Park, NJ  
🌐 [LinkedIn](#) 📄 [Google Scholar](#) 🏠 [GitHub](#)

## PROFESSIONAL SUMMARY

Ph.D. candidate developing interpretable statistical methods for high-dimensional time-series data. Research centers on identifiable dynamic factor models, sparse factor transitions, Bayesian inference, and state-space estimation, with applications to environmental and economic forecasting. Approximately three years of prior quantitative-finance experience in derivatives, forecasting, and risk analysis.

## EDUCATION

### Rutgers University

New Brunswick, NJ

*Doctor of Philosophy (PhD) in Statistics*

Sep 2022 – Present

- ▷ GPA: 3.84/4.0 | Advisors: Prof. Gemma Moran, Prof. Han Xiao
- ▷ Research: Dynamic factor models, sparse inference, deep generative models
- ▷ MS in Statistics awarded May 2024 (en route to PhD)
- ▷ Advanced to doctoral candidacy (Nov 2025); dissertation proposal approved by committee

### James Madison University

Harrisonburg, VA

*BS Mathematics & Economics (Double Major)*

Jan 2016 – May 2019

- ▷ **Valedictorian**, College of Business | GPA: 3.95/4.0 | Phi Beta Kappa

## ACADEMIC EXPERIENCE

### Research Assistant

Sep 2022 – Present

*Department of Statistics, Rutgers University*

New Brunswick, NJ

- ▷ Conducted research on enhancing time series analysis with machine learning algorithms, focusing on temporal trends and seasonal pattern modeling
- ▷ Applied deep learning techniques (e.g., LSTMs, VAEs) for predictive modeling, improving the robustness and accuracy of forecasts
- ▷ Collaborated with cross-disciplinary teams to integrate statistical theories with practical machine learning applications; contributed to peer-reviewed articles on deep learning applications, including retinal abnormality detection and stock market prediction
- ▷ Dissertation research: Dynamic Vector Factor Model with Sparse Factor Transitions – Bayesian dynamic factor model with spike-and-slab priors on factor-innovation variances; EM estimation with Kalman filtering and RTS smoothing (advisors: Prof. Gemma Moran, Prof. Han Xiao)
- ▷ Co-author, stochastic weather generators for high-frequency wind vector time series (with G. E. Moran and M. L. Stein): stochastic-volatility validation and quantile-regression analysis; arXiv:2606.09941, under review at ASCMO

### Statistical Consultant, Office of Statistical Consulting (OSC)

Sep 2023 – Present

*Department of Statistics, Rutgers University*

New Brunswick, NJ

- ▷ Provide statistical consulting to Rutgers faculty, researchers, and graduate students under faculty supervision: study design, power analysis, data analysis, and interpretation of results
- ▷ Advise on statistical methodology for grant proposals and manuscripts across disciplines

### Teaching Assistant

Sep 2022 – Present

*Department of Statistics, Rutgers University*

New Brunswick, NJ

- ▷ **Best Teaching Assistant Award (2024)** – Statistical Inference, Regression, Time Series, ML
- ▷ Led undergraduate lectures in topics like hypothesis testing and linear regression, improving class average scores
- ▷ Hosted weekly recitation sessions using Python and R, bolstering students' computational skills
- ▷ Assisted instructors in exam creation, grading, and feedback, ensuring pedagogical objectives were met

### Research Assistant

Aug 2017 – Dec 2018

*Department of Computer Information Systems, James Madison University*

Harrisonburg, VA

- ▷ Collected and organized healthcare delivery data for Fredericksburg hospitals

- ▷ Automated data processing tasks with Excel and R, significantly reducing the time required for complex data manipulations and analysis
- ▷ Conducted time series analysis on healthcare data using ARIMA, VAR, and multivariate methods to uncover patterns and predict trends in hospital delivery systems

## RESEARCH & PUBLICATIONS

---

### Research in Progress:

- ▷ **Dynamic Vector Factor Model with Sparse Factor Transitions** – Novel Linear-Gaussian DFM achieving identifiability through episodic factor activation; MAP-EM estimation with Kalman filtering and RTS smoothing.

### Preprints:

- ▷ Cui M, Eng K, Greene JT, Ke Z, **Sodagartojgi A**, Xia Z, Moran GE, Stein ML (2026). Stochastic weather generators for high-frequency wind vector time series. [arXiv:2606.09941](https://arxiv.org/abs/2606.09941). Under review at *Advances in Statistical Climatology, Meteorology and Oceanography* (ASCMO), ms. ascmo-2026-28

### Peer-Reviewed Publications:

#### *Statistical modeling and machine learning:*

- ▷ Raeisi Z, **Sodagartojgi A**, Sharafkhani F, Roshanzamir A, Najafzadeh H, Bashiri O, Golkarieh A (2025). Enhanced classification of tinnitus patients using EEG microstates and deep learning techniques. *Scientific Reports*, 15(1), 15959 [28 citations]
- ▷ Ahmadi Lashaki R, Raeisi Z, **Sodagartojgi A**, Abedi Lomer F, Aghdaei E, Najafzadeh H (2025). EEG microstate analysis in trigeminal neuralgia: identifying potential biomarkers for enhanced diagnostic accuracy. *Acta Neurologica Belgica*, 125(4), 1025–1045 [12 citations]
- ▷ Talebzadeh M, **Sodagartojgi A**, Moslemi Z, Sedighi S, Kazemi B, Akbari F (2024). Deep learning-based retinal abnormality detection from OCT images with limited data. *World Journal of Advanced Research and Reviews*, 21(3), 690–698 [23 citations]

#### *Interdisciplinary collaborations — applied modeling and optimization:*

- ▷ Dokhanian S, **Sodagartojgi A**, Tehranian K, Ahmadirad Z, Moghaddam PK, Mohsenibeigzadeh M (2024). Exploring the impact of supply chain integration and agility on commodity supply chain performance. *World Journal of Advanced Research and Reviews*, 22(1), 441–450 [40 citations]
- ▷ Tashakkori A, Erfanibehrouz N, Mirshekari S, **Sodagartojgi A**, Gupta V (2024). Enhancing stock market prediction accuracy with recurrent deep learning models: a case study on the CAC40 index. *World Journal of Advanced Research and Reviews*, 23(1), 2309 [22 citations]
- ▷ Abdollahi SA, Pourabadeh A, Alishiri M, **Sodagartojgi A**, Ranjbar SF, Ehghaghi MB, Talati F (2024). Modeling and optimization of efficient removal of diclofenac and naproxen based on chemometric approaches. *Water Resources and Industry*, 31, 100238 [16 citations]
- ▷ Heidari S, Shafiesabet A, Darvishan S, Karimi R, **Sodagartojgi A** (2026). Modeling the relationship between artificial intelligence, service innovation, and financial performance: a dual SEM–ANN perspective. *International Journal of Research in Industrial Engineering*, 15(3), 576–594

## PRESENTATIONS & PROFESSIONAL SERVICE

---

- ▷ **Poster presentation:** Dynamic Factor Model with Sparse Factor Transitions. *2025 NBER–NSF Time Series Conference*, Rutgers University, New Brunswick, NJ, September 19–20, 2025
- ▷ **Peer reviewer**, *Computer Methods in Biomechanics and Biomedical Engineering* (Taylor & Francis), 2026
- ▷ **Conference organization:** assisted with organizing *The Design and Analysis of Experiments (DAE) Conference 2026*, Rutgers University, Piscataway, NJ, May 19–21, 2026

# PROFESSIONAL EXPERIENCE

---

## Commodity Derivatives Trading Analyst

Oct 2020 – Jul 2022

*Rabobank*

New York, NY

- ▷ Managed energy and agriculture trading portfolios using quantitative models to analyze commodity hedges and derivatives
- ▷ Developed machine learning models to forecast market trends, integrating time series data with predictive analytics to enhance trading strategies
- ▷ Designed and implemented Excel VBA-based tools for real-time risk management and pricing analysis
- ▷ Executed pricing strategies for a diverse range of financial derivatives, including dairy and heating oil options
- ▷ Advanced trading acumen, executing and analyzing commodity hedges using futures, options, and structured products
- ▷ Generated comprehensive end-of-day reports to assess commodity position health, informing strategy adjustments

## Corporate Derivatives Sales Analyst

Aug 2019 – Mar 2021

*Rabobank*

New York, NY

- ▷ Collaborated with cross-functional teams to conduct research-driven risk assessments, using statistical models and market data to evaluate and manage economic exposures, ensuring clients' positions were strategically aligned with their financial goals
- ▷ Engineered financial hedging models for international clients, incorporating interest rate swaps and structured products
- ▷ Built real-time dashboards for commodity trading, integrating machine learning algorithms to predict price movements in volatile markets
- ▷ Conducted sophisticated analysis and devised comprehensive models for currencies and interest rates, which were instrumental in guiding clients' financial strategies and managing economic exposures

# TECHNICAL SKILLS

---

**Programming:** Python, R, MATLAB, SQL, C++, Julia, SAS

**ML/DL:** PyTorch, TensorFlow, JAX, scikit-learn, XGBoost

**Methods:** Bayesian Inference, State-Space Models, Causal Inference, VAEs, Transformers

**Finance:** Derivatives Pricing, Risk Management, FINRA Series 7/79/63, Certified FRM

# HONORS & AWARDS

---

- ▷ Fully funded PhD appointment, Rutgers (2022–2027)
- ▷ Best Teaching Assistant Award, Rutgers (2024)
- ▷ Best Qualifying Exam Performance Award (written exams), Rutgers (2023)
- ▷ College of Business Valedictorian, JMU (2019)
- ▷ Scholarship of Distinction, JMU
- ▷ CGE Scholarship, JMU
- ▷ Phi Beta Kappa Honor Society (2019)
- ▷ Outstanding Mathematics Student, JMU (2019)